

10

Making measurements in relativity

10.1 Observers and their observations	108
10.2 Coordinate and non-coordinate bases	110
10.3 The orthonormal frame	114
10.4 Freely falling frames	116
Chapter summary	118
Exercises	118

Why were another seven years required for the construction of the general theory of relativity? The main reason lies in the fact that it is not easy to free oneself from the idea that coordinates must have an immediate metrical meaning. Einstein quoted in P. A. Schilpp (ed.) *Albert Einstein – Philosopher Scientist* (1969).

The stage on which the drama of general relativity is played out is curved spacetime. However, measurements are made *locally* by observers in laboratories. Over the small distances involved in a typical experiment, an observer will experience spacetime as if it were **flat spacetime** with the Minkowski metric. We therefore need to know how to relate the observations made by observers in their local spacetime to the objects we manipulate in the curved spacetime of general relativity. The key is that measurements are made in local orthonormal frames: frames of reference set up by observers where the basis vectors are usually orthogonal and normalized. In this book, component labels in such frames will be given with hats, so that the basis vectors, for example, will be written as $e_{\hat{\alpha}}$. By definition, the metric of the flat, local frame is simply the Minkowski metric, with components $\eta_{\hat{\mu}\hat{\nu}} = e_{\hat{\mu}} \cdot e_{\hat{\nu}} = \text{diag}(-1, 1, 1, 1)$.

A further point to note in these discussions stems from Einstein's observation in the quotation above. When presented with a t coordinate or an r coordinate, it is tempting to assume that t must represent time and r the radius. This is not correct. Coordinates are intrinsically meaningless labels which are only given meaning by relating them to measurements and intervals determined by observers in their local, inertial frames of reference. This can be summed up using the slogan that *coordinates have no immediate metrical significance*.¹

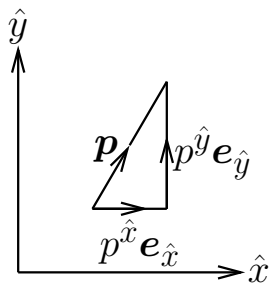


Fig. 10.1 A measurement of momentum $\mathbf{p}$. In this figure, we use the familiar $\hat{x}$ and $\hat{y}$ axes of two-dimensional space (with basis vectors $e_{\hat{x}}$ and $e_{\hat{y}}$), but the idea carries over to the four-dimensional spacetime axes of an orthonormal frame with basis vectors $e_{\hat{0}}, e_{\hat{1}}, e_{\hat{2}}$ and $e_{\hat{3}}$.

10.1 Observers and their observations

A particle passes through a laboratory as shown in Fig. 10.1. The particle has a momentum $\mathbf{p}$ which, expressed as a vector, is a quantity independent of any set of coordinates. An observer in the laboratory makes measurements by carrying around their own orthonormal axes $e_{\hat{0}}, e_{\hat{1}}, e_{\hat{2}}$ and $e_{\hat{3}}$. Referred to these axes, the vector can be expressed in coordinates as $\mathbf{p} = p^{\hat{\mu}} e_{\hat{\mu}}$. To make a measurement of a particular

component of a vector the observer projects out the component using their local axes. For example, measuring the momentum along the $\hat{\alpha}$ direction means that the observer makes the projection via a dot product $\mathbf{p} \cdot \mathbf{e}_{\hat{\alpha}}$.

Example 10.1

The observer makes the projection

$$\begin{aligned}\mathbf{p} \cdot \mathbf{e}_{\hat{\alpha}} &= p^{\hat{\mu}} \mathbf{e}_{\hat{\mu}} \cdot \mathbf{e}_{\hat{\alpha}} \\ &= p^{\hat{\mu}} \eta_{\hat{\mu}\hat{\alpha}} = p_{\hat{\alpha}},\end{aligned}\quad (10.1)$$

showing that the observer has access to the component $p_{\hat{\alpha}}$. The observer can use $\eta^{\hat{\alpha}\hat{\beta}}$ to raise the index, if they want the up-index form of the component via $p^{\hat{\beta}} = \eta^{\hat{\alpha}\hat{\beta}} p_{\hat{\alpha}}$.

If we spot an observer, how do we know what their local orthonormal axes will look like? That is, how will they orient their orthonormal coordinates? Start by noting that the observer's world line is characterized by their velocity vector $\mathbf{u}_{\text{obs}}$, which is tangent to the world line (Fig. 10.2). The key is that the timelike vector of the local basis $\mathbf{e}_{\hat{0}}$ will also be tangent to the observer's world line, since this is the direction that a clock at rest in the observer's frame moves in spacetime. We therefore have

$$\mathbf{e}_{\hat{0}} = \mathbf{u}_{\text{obs}}. \quad (10.2)$$

Therefore, expressed in some coordinate frame, the observer's timelike axis $\mathbf{e}_{\hat{0}}$ has the components we would ascribe to their tangent vector $\mathbf{u}_{\text{obs}}$. We write these components of the observer's timelike basis vector

$$(\mathbf{e}_{\hat{0}})^{\mu} = u_{\text{obs}}^{\mu}. \quad (10.3)$$

The other components of the observer's orthonormal system can then be picked out, subject to being orthogonal to $\mathbf{e}_{\hat{0}}$ and to each other.

Example 10.2

Consider the constantly accelerated observer in Minkowski space from Chapter 2. They have a timelike basis vector with components

$$(\mathbf{e}_{\hat{0}})^{\mu} = u_{\text{obs}}^{\mu}(\tau) = (\cosh(g\tau), \sinh(g\tau), 0, 0). \quad (10.4)$$

Pick $\mathbf{e}_{\hat{2}}$ and $\mathbf{e}_{\hat{3}}$ to point along the y - and z -directions. The remaining 4-vector $\mathbf{e}_{\hat{1}}$ has the form $(f(\tau), g(\tau), 0, 0)$. We require orthogonality of the observer's basis vectors, which is to say that

$$\eta_{\mu\nu} (\mathbf{e}_{\hat{0}})^{\mu} (\mathbf{e}_{\hat{1}})^{\nu} = -\cosh(g\tau)f(\tau) + \sinh(g\tau)g(\tau) = 0. \quad (10.5)$$

We also require that the vectors are normalized in the observer's frame, so that

$$-f^2(\tau) + g^2(\tau) = 1. \quad (10.6)$$

This allows us to solve for $(\mathbf{e}_{\hat{1}})^{\mu}$ and we end up with

$$\begin{aligned}(\mathbf{e}_{\hat{0}})^{\mu} &= (\cosh(g\tau), \sinh(g\tau), 0, 0), \\ (\mathbf{e}_{\hat{1}})^{\mu} &= (\sinh(g\tau), \cosh(g\tau), 0, 0), \\ (\mathbf{e}_{\hat{2}})^{\mu} &= (0, 0, 1, 0), \\ (\mathbf{e}_{\hat{3}})^{\mu} &= (0, 0, 0, 1).\end{aligned}\quad (10.7)$$

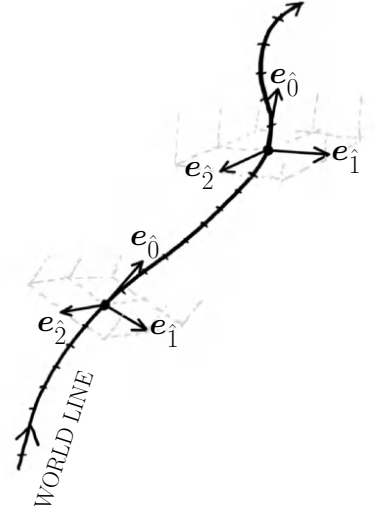


Fig. 10.2 Local orthonormal frames picked out along a world line by setting $\mathbf{e}_{\hat{0}} = \mathbf{u}$.

One particularly helpful tool is that the energy of a particle measured by an observer with velocity $\mathbf{u}_{\text{obs}}$ is given by $E = -\mathbf{p} \cdot \mathbf{u}_{\text{obs}}$. This is easily confirmed by noting that

$$\begin{aligned} -\mathbf{p} \cdot \mathbf{u}_{\text{obs}} &= -p^{\hat{\mu}} \mathbf{e}_{\hat{\mu}} \cdot \mathbf{e}_{\hat{0}} \\ &= -p^{\hat{\mu}} \eta_{\hat{\mu}0} = -p_{\hat{0}} = E, \end{aligned} \quad (10.8)$$

where, in the final step we remember that $E = p^{\hat{0}} = -p_{\hat{0}}$ because $\eta_{\hat{0}\hat{0}} = -1$ in the orthonormal frame.

Example 10.3

Consider Minkowski space. In a frame where a particle is at rest, the particle has $p^{\mu} = (m, 0, 0, 0)$. Relative to this frame the observer travels with constant speed v along the x -axis, so the 4-velocity of the observer has components $u_{\text{obs}}^{\mu} = (\gamma, \gamma v, 0, 0)$, which are therefore also the components of the local basis vector $(\mathbf{e}_{\hat{0}})^{\mu}$. The energy of the particle measured by the observer, when the world lines of particle and observer intersect, is²

$$E = -\mathbf{p} \cdot \mathbf{u}_{\text{obs}} = m\gamma. \quad (10.9)$$

So the particle has energy $E = \gamma mc^2$ (restoring factors of c) as we expect.

Now consider the accelerated observer from the previous example, measuring light from a star in Minkowski space. The star gives out light at a frequency ω . The wave 4-vector of a photon reaching the observer has components³ $k^{\mu} = (\omega, \omega, 0, 0)$ in the star's rest frame. The method for finding the frequency the observer measures is the same as for the energy, to which frequency is proportional. We evaluate $\omega = -\mathbf{k} \cdot \mathbf{u}_{\text{obs}}$. Computing, we find

$$\begin{aligned} \omega(\tau) &= -\mathbf{k} \cdot \mathbf{u}_{\text{obs}} \\ &= k^0 u^0 - k^1 u^1 \\ &= \omega [\cosh(g\tau) - \sinh(g\tau)] \\ &= \omega \exp(-g\tau), \end{aligned} \quad (10.10)$$

which demonstrates that the shift in observed frequency varies exponentially.⁴

The procedure above allows us to understand what an observer will measure. However, this is complicated by the fact that the natural, orthonormal coordinate system that observers employ has an unpleasant mathematical property: it is a **non-coordinate basis**, as we now describe.

10.2 Coordinate and non-coordinate bases

Recall from Chapter 3, that plane polar coordinates had the property that $|\mathbf{e}_r| = 1$ but that $|\mathbf{e}_{\theta}| = r$. That is, the length of the $\mathbf{e}_{\theta}$ basis vector is proportional to the distance away from the origin. This coordinate system was derived from the Cartesian one by expressing components as partial derivatives with respect to the Cartesian coordinates. We call such a coordinate system a **coordinate basis**.

We could choose to **normalize** $\mathbf{e}_{\theta}$ so that we have $\mathbf{e}_{\hat{\theta}} = \mathbf{e}_{\theta}/r$, which yields an orthonormal basis set $\mathbf{e}_{\hat{r}} (= \mathbf{e}_r)$ and $\mathbf{e}_{\hat{\theta}}$. Although this is the

²We can evaluate the dot product using the Minkowski tensor in the rest frame of the particle. The result is a scalar, so is true in any frame.

³In our units, the vector $\mathbf{k}$ has components $k^{\mu} = (\omega, k^x, k^y, k^z)$ and $\omega = |\vec{k}|$ for light. We also assume the quantum mechanical relationship $E = \hbar\omega$, but set $\hbar = 1$.

⁴This result, discussed in the book by Hartle, will be seen again in the discussion of black holes in Chapters 26 and 27.

↪ The rest of this chapter goes through further detail on coordinates and their bases, and how to transform between them. A reader impatient to get on to the topic of curvature may skip to the start of the next chapter on a first reading.

coordinate set we would most probably want to choose when plotting the position of events in the laboratory, it does not have the property that it is derivable directly from the Cartesian coordinates. That is to say that the basis vectors $\mathbf{e}_r$ and $\mathbf{e}_\theta$ cannot be written as an expansion in Cartesian basis vectors with prefactors given in terms of derivatives of to the Cartesian components with respect to r and θ (see eqn 3.7). We call such a basis a⁵ **non-coordinate basis**.

⁵As described in Chapter 3, a non-coordinate basis can be identified because its basis vectors don't commute, in contrast to the basis vectors of a coordinate basis which do commute.

Example 10.4

Non-coordinate bases are useful for many problems. Consider, for example, the Kepler problem, which is conventionally discussed using an orthonormal basis and cylindrical coordinates. In such a frame, the velocity $\mathbf{v} = v^{\hat{r}}\mathbf{e}_{\hat{r}} + v^{\hat{\theta}}\mathbf{e}_{\hat{\theta}}$ is given by

$$\mathbf{v} = \frac{dr}{dt}\mathbf{e}_{\hat{r}} + r\frac{d\mathbf{e}_{\hat{r}}}{dt}, \quad (10.11)$$

that is, we take a time derivative of $\mathbf{r} = r\mathbf{e}_{\hat{r}}$. Acceleration is then given by

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{dv^{\hat{r}}}{dt}\mathbf{e}_{\hat{r}} + v^{\hat{r}}\frac{d\mathbf{e}_{\hat{r}}}{dt} + \frac{dv^{\hat{\theta}}}{dt}\mathbf{e}_{\hat{\theta}} + v^{\hat{\theta}}\frac{d\mathbf{e}_{\hat{\theta}}}{dt}. \quad (10.12)$$

The basis vectors, as shown in Fig. 10.3, obey

$$\frac{d\mathbf{e}_{\hat{r}}}{dt} = \frac{d\theta}{dt}\mathbf{e}_{\hat{\theta}} = \omega\mathbf{e}_{\hat{\theta}}, \quad (10.13)$$

$$\frac{d\mathbf{e}_{\hat{\theta}}}{dt} = -\frac{d\theta}{dt}\mathbf{e}_{\hat{r}} = -\omega\mathbf{e}_{\hat{r}}, \quad (10.14)$$

where $\omega = d\theta/dt$. The components of the acceleration are then written as

$$a^{\hat{r}} = \frac{dv^{\hat{r}}}{dt} - v^{\hat{\theta}}\frac{d\theta}{dt} = \frac{d^2r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2 = \ddot{r} - r\dot{\theta}^2, \quad (10.15)$$

and

$$a^{\hat{\theta}} = \frac{dv^{\hat{\theta}}}{dt} + v^{\hat{r}}\frac{d\theta}{dt} = \frac{d}{dt}\left(r\frac{d\theta}{dt}\right) + \frac{dr}{dt}\frac{d\theta}{dt} = r\ddot{\theta} + 2\dot{r}\dot{\theta}. \quad (10.16)$$

We shall usually work in coordinate bases owing to their neat geometrical properties. However, observers work in local frames, which is where their measurements are made. These are chosen to be orthonormal and hence usually have non-coordinate bases. We therefore need to be able to transform between coordinate bases (where we do our calculations) and non-coordinate bases (where measurements are made).

The basis vectors in a coordinate basis are related to the metric via the definition

$$\mathbf{e}_\mu \cdot \mathbf{e}_\nu = g_{\mu\nu}. \quad (10.17)$$

Measurements are made in the local orthonormal frame, with Minkowski metric whose components are related to the dot product of the basis vectors via

$$\mathbf{e}_{\hat{\alpha}} \cdot \mathbf{e}_{\hat{\beta}} = \eta_{\hat{\alpha}\hat{\beta}}. \quad (10.18)$$

Any vector can be decomposed in either system:

$$\mathbf{a} = a^\alpha \mathbf{e}_\alpha = a^{\hat{\beta}} \mathbf{e}_{\hat{\beta}}. \quad (10.19)$$

↪ See Chapter 20 for a discussion of the Kepler problem.

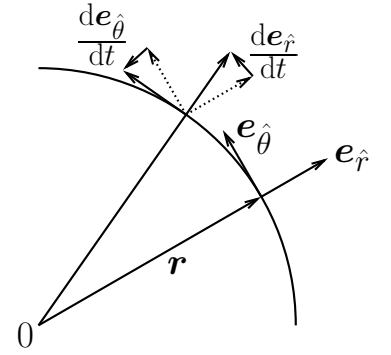


Fig. 10.3 The change of the vector $\mathbf{e}_{\hat{r}}$ is in the direction $\mathbf{e}_{\hat{\theta}}$, which the change in $\mathbf{e}_{\hat{\theta}}$ is in the direction $-\mathbf{e}_{\hat{r}}$.

In order to transform between these two descriptions, we write the components of the orthonormal basis vectors in the coordinate frame as $(\mathbf{e}_{\hat{\beta}})^\alpha$, giving us an expression

$$a^\alpha = a^{\hat{\beta}}(\mathbf{e}_{\hat{\beta}})^\alpha. \quad (10.20)$$

Objects such as $(\mathbf{e}_{\hat{\beta}})^\alpha$ are a set of matrices known as the components of a **vielbein**.⁶ These turn out to be very useful.⁷ We can also write the components of the coordinate basis vectors in the orthonormal frame, which leads to the expression

$$a^{\hat{\beta}} = a^\alpha(\mathbf{e}_\alpha)^{\hat{\beta}}. \quad (10.21)$$

⁶Which translates from German into English as many-leg. Since the vielbein with which we're concerned describes (3+1)-dimensional spacetime, it is sometimes called a vierbein ($\equiv$ 4-leg). The bracket in the notation $(\mathbf{e}_\mu)^\alpha$ is really just there for aesthetic reasons to remind us that a vielbein combines information about two different sorts of coordinate systems. We could write the components e_μ^α if we prefer.

⁷One way to think of the action of a vielbein is that the coordinate frame possesses a set of global coordinates [e.g. (t, r, θ, ϕ)] and that we make them local using the vielbein.

Example 10.5

Consider cylindrical-polar coordinates. The coordinate basis consists of vectors $\mathbf{e}_r$ and $\mathbf{e}_\theta$ and we write components (r, θ) . The metric is given by the line element

$$ds^2 = dr^2 + r^2 d\phi^2. \quad (10.22)$$

The off-diagonal elements of the metric are zero; while the diagonal components are $g_{rr} = 1$ and $g_{\theta\theta} = r^2$. From $g_{\mu\nu} = \mathbf{e}_\mu \cdot \mathbf{e}_\nu$, the coordinate basis vectors obey

$$|\mathbf{e}_r| = 1, \quad |\mathbf{e}_\theta| = r, \quad (10.23)$$

and we argued that a good set of orthonormal basis vectors are given by

$$\mathbf{e}_{\hat{r}} = \mathbf{e}_r, \quad \mathbf{e}_{\hat{\theta}} = \frac{1}{r} \mathbf{e}_\theta. \quad (10.24)$$

Let's first look at how vielbein notation works. Quite trivially, we can say that the components of the coordinate basis vectors in the coordinate basis are, by definition

$$(\mathbf{e}_r)^\mu = (1, 0), \quad (\mathbf{e}_\theta)^\mu = (0, 1). \quad (10.25)$$

Similarly, in the orthonormal frame we write the trivial equation

$$(\mathbf{e}_{\hat{r}})^{\hat{\mu}} = (1, 0), \quad (\mathbf{e}_{\hat{\theta}})^{\hat{\mu}} = (0, 1). \quad (10.26)$$

More interestingly, the components of the orthonormal basis vectors in the coordinate basis can be written down and are⁸

$$(\mathbf{e}_{\hat{r}})^\mu = (1, 0), \quad (\mathbf{e}_{\hat{\theta}})^\mu = (0, 1/r). \quad (10.27)$$

How do we know that the orthonormal vectors we have selected are correct? The key is that they must obey the defining relationship $\eta_{\hat{\mu}\hat{\nu}} = \mathbf{e}_{\hat{\mu}} \cdot \mathbf{e}_{\hat{\nu}} = g_{\alpha\beta}(\mathbf{e}_{\hat{\mu}})^\alpha(\mathbf{e}_{\hat{\nu}})^\beta$ and this is quickly checked using $g_{rr} = 1$ and $g_{\theta\theta} = r^2$. For example

$$\begin{aligned} \mathbf{e}_{\hat{r}} \cdot \mathbf{e}_{\hat{r}} &= g_{rr}(\mathbf{e}_{\hat{r}})^r(\mathbf{e}_{\hat{r}})^r + g_{\theta\theta}(\mathbf{e}_{\hat{r}})^\theta(\mathbf{e}_{\hat{r}})^\theta = 1 + 0 = 1, \\ \mathbf{e}_{\hat{\theta}} \cdot \mathbf{e}_{\hat{\theta}} &= g_{rr}(\mathbf{e}_{\hat{\theta}})^r(\mathbf{e}_{\hat{\theta}})^r + g_{\theta\theta}(\mathbf{e}_{\hat{\theta}})^\theta(\mathbf{e}_{\hat{\theta}})^\theta = 0 + r^2 \frac{1}{r} \frac{1}{r} = 1. \end{aligned} \quad (10.28)$$

This shows that our choice is correct, since $\eta_{\hat{r}\hat{r}} = \eta_{\hat{\theta}\hat{\theta}} = 1$. Similarly, it follows that the coordinate basis vectors in the orthonormal basis are

$$(\mathbf{e}_r)^{\hat{\mu}} = (1, 0), \quad (\mathbf{e}_\theta)^{\hat{\mu}} = (0, r). \quad (10.29)$$

These must obey the defining relationship $\mathbf{e}_\mu(x) \cdot \mathbf{e}_\nu(x) = g_{\mu\nu}(x)$, which they do.

A vielbein can be presented as a matrix as we now demonstrate.

Example 10.6

Consider the metric with line element $ds^2 = d\theta^2 + \sin^2\theta d\phi^2$. Using the method above, we can compute the matrix representing the components of the coordinate basis vectors in the orthonormal basis⁹

$$(e_\mu)^{\hat{\alpha}} = \begin{pmatrix} (e_1)^{\hat{1}} & (e_1)^{\hat{2}} \\ (e_2)^{\hat{1}} & (e_2)^{\hat{2}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & \sin\theta \end{pmatrix}. \quad (10.30)$$

As the vielbein is simply a square matrix, the usual rules for inverses apply and so we have

$$(e_\mu)^{\hat{\alpha}}(e_{\hat{\alpha}})^\nu = \delta^\nu_\mu \quad (e_\mu)^{\hat{\alpha}}(e_{\hat{\beta}})^\mu = \delta^{\hat{\alpha}}_{\hat{\beta}}. \quad (10.31)$$

The inverse matrix follows representing the components of the orthonormal basis vectors in the coordinate basis

$$(e_{\hat{\alpha}})^\mu = \begin{pmatrix} (e_1)^1 & (e_1)^2 \\ (e_2)^1 & (e_2)^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{\sin\theta} \end{pmatrix}. \quad (10.32)$$

Once we have the vielbein we need the general rule¹⁰ that the vielbein components $(e_{\hat{\mu}})^\beta$ remove a down coordinate component β and replaces it with an orthonormal component $\hat{\mu}$. It also replaces the up component p^μ with component p^β . That is to say

$$(e_{\hat{\mu}})^\beta p_\beta = p_{\hat{\mu}}, \quad (e_{\hat{\mu}})^\beta p^{\hat{\mu}} = p^\beta. \quad (10.38)$$

We also have

$$(e_\mu)^{\hat{\beta}} p_{\hat{\beta}} = p_\mu, \quad (e_\mu)^{\hat{\beta}} p^{\hat{\beta}} = p^\beta. \quad (10.39)$$

These equations also apply to tensor components, so we might have, for example

$$T_{\alpha\beta} = (e_\alpha)^{\hat{\mu}}(e_\beta)^{\hat{\nu}} T_{\hat{\mu}\hat{\nu}}. \quad (10.40)$$

To summarize, a vielbein allows us to set up orthonormal frames across all spacetime according to the defining rules

$$g_{\mu\nu} = \eta_{\hat{\alpha}\hat{\beta}}(e_\mu)^{\hat{\alpha}}(e_\nu)^{\hat{\beta}}, \quad (10.41)$$

$$\eta_{\hat{\alpha}\hat{\beta}} = g_{\mu\nu}(e_{\hat{\alpha}})^\mu(e_{\hat{\beta}})^\nu. \quad (10.42)$$

Example 10.7

The use of vielbein components generalizes the rule that energy measured by an observer with velocity $\mathbf{u}_{\text{obs}}$ is given by $E = -\mathbf{p} \cdot \mathbf{u}_{\text{obs}}$, where $\mathbf{p}$ is the momentum vector. This is because we always choose $\mathbf{e}_{\hat{0}} = \mathbf{u}_{\text{obs}}$. To prove this we write

$$E = -g_{\mu\nu}p^\mu(e_{\hat{0}})^\nu = -p_\nu(e_{\hat{0}})^\nu. \quad (10.43)$$

We know from the definitions of how a vielbein works that $-p_\nu(e_{\hat{0}})^\nu = -p_{\hat{0}}$. Finally, since in the orthonormal frame, indices are manipulated with the Minkowski tensor η , we have that $E = -p_{\hat{0}} = -\eta_{\hat{0}\hat{0}}p^{\hat{0}} = p^{\hat{0}}$, as we require.

⁹ The vielbein components in this case can be written as

$$(e_\theta)^{\hat{\theta}} = 1, \quad (e_\phi)^{\hat{\phi}} = \sin\theta.$$

¹⁰Life is made easier in understanding the action of a vielbein if we also employ our knowledge of 1-forms. Formally we define the action of the vielbein on basis vectors and basis 1-forms via

$$\begin{aligned} e_{\hat{\mu}} &= (e_{\hat{\mu}})^\alpha e_\alpha, & \omega^{\hat{\mu}} &= (e_\alpha)^{\hat{\mu}} \omega^\alpha \\ e_\alpha &= (e_\alpha)^{\hat{\mu}} e_{\hat{\mu}}, & \omega^\alpha &= (e_{\hat{\mu}})^\alpha \omega^{\hat{\mu}}. \end{aligned} \quad (10.33)$$

From which the rules in the text can be confirmed. Equivalently, we can return to the definition

$$\mathbf{v} = v^\mu \mathbf{e}_\mu = v^{\hat{\nu}} \mathbf{e}_{\hat{\nu}}, \quad (10.34)$$

and use the inner product $\langle \omega^\alpha, \mathbf{e}_\mu \rangle = \delta^\alpha_\mu$. To remove a vector like $\mathbf{e}_\mu$, from the second term in eqn 10.34, we note that, on taking an inner product with basis 1-form ω^μ , we have

$$v^\mu = v^{\hat{\nu}} \langle \omega^\mu, \mathbf{e}_{\hat{\nu}} \rangle = v^{\hat{\nu}} (e_{\hat{\nu}})^\mu, \quad (10.35)$$

where we've fixed the vielbein components by

$$\langle \omega^\mu, \mathbf{e}_{\hat{\nu}} \rangle = \langle \omega^\mu, \mathbf{e}_\alpha \rangle (e_{\hat{\nu}})^\alpha = (e_{\hat{\nu}})^\mu. \quad (10.36)$$

Note here that $(e_{\hat{\nu}})^\mu = \langle \omega^\mu, \mathbf{e}_{\hat{\nu}} \rangle \neq \delta^\mu_{\hat{\nu}}$, since here we're working with the basis vectors of two different coordinate systems. The same method also yields

$$v^{\hat{\mu}} = v^\nu \langle \omega^{\hat{\mu}}, \mathbf{e}_\nu \rangle = v^\nu (e_\nu)^{\hat{\mu}}. \quad (10.37)$$

The other relationships can be confirmed using the idea that a 1-form can be written as $\tilde{\mathbf{u}} = u_\mu \omega^\mu = u_{\hat{\mu}} \omega^{\hat{\mu}}$.

Vielbein components will be very useful to us. Next, we need to identify some examples of frames in which observers make their measurements.

10.3 The orthonormal frame

In many of the cases we'll consider later in the book, we shall find that a very convenient orthonormal frame in which to carry out computations, particularly of curvature, is one where the observer is at rest relative to the coordinate frame. That is, the velocity expressed in the coordinate frame is $u^\mu = dx^\mu/d\tau = (u^0, 0, 0, 0)$, with u^0 fixed such that $g_{\mu\nu}u^\mu u^\nu = g_{00}(u^0)^2 = -1$. This observer then chooses $e_{\hat{t}} = \mathbf{u}$. We will call this rather natural choice¹¹ **the stationary orthonormal frame**. In such a frame, we have a metric that looks locally like the Minkowski metric, but there is no reason to believe that the connection coefficients should vanish.¹²

To find the orthonormal frame we effectively diagonalize the metric and normalize the components. That is, we are trying to solve

$$g(e_{\hat{\alpha}}, e_{\hat{\beta}}) = \eta_{\hat{\alpha}\hat{\beta}}, \quad (10.44)$$

which is equivalent to the component equation

$$g_{\mu\nu}(e_{\hat{\alpha}})^\mu (e_{\hat{\beta}})^\nu = \eta_{\hat{\alpha}\hat{\beta}}. \quad (10.45)$$

¹¹Later in the book, if we don't specify that the orthonormal frame is that of a specific observer, one can assume we're referring to the stationary orthonormal frame of an observer who is stationary in the coordinate frame.

¹²It is a local inertial frame (LIF) where the connection coefficients vanish at the point in question. This more restrictive condition is dealt with in the next section.

Example 10.8

The matrix form for (1+1)-dimensional spacetime is

$$\begin{pmatrix} (e_{\hat{0}})^0 & (e_{\hat{0}})^1 \\ (e_{\hat{1}})^0 & (e_{\hat{1}})^1 \end{pmatrix} \begin{pmatrix} g_{00} & g_{01} \\ g_{10} & g_{11} \end{pmatrix} \begin{pmatrix} (e_{\hat{0}})^0 & (e_{\hat{1}})^0 \\ (e_{\hat{0}})^1 & (e_{\hat{1}})^1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (10.46)$$

In the most commonly encountered case of a metric that is already diagonal, we can simply normalize the components as discussed in the next example.

Example 10.9

For an observer at rest $\mathbf{u}$ has a single non-zero component u^0 . Its value is given via the normalization of the velocity, by

$$g_{00}(u^0)^2 = -1, \quad (10.47)$$

so we must have $u^0 = (-g_{00})^{-1/2}$. Our rule $e_{\hat{0}} = \mathbf{u}$ then mandates $e_{\hat{0}} = e_0/\sqrt{-g_{00}}$. For the diagonal metric we can then pick out an orthonormal basis

$$e_{\hat{0}} = \frac{1}{\sqrt{-g_{00}}}e_0, \quad e_{\hat{i}} = \frac{1}{\sqrt{g_{ii}}}e_i, \quad (10.48)$$

with components in the coordinate frame of

$$(e_{\hat{0}})^0 = \frac{1}{\sqrt{-g_{00}}}, \quad (e_{\hat{i}})^i = \frac{1}{\sqrt{g_{ii}}}. \quad (10.49)$$

We can check this works by considering the rule $g_{\mu\nu}(e_{\hat{\alpha}})^\mu (e_{\hat{\beta}})^\nu = \eta_{\hat{\alpha}\hat{\beta}}$. If the metric g is diagonal then, by inspection, we have

$$\begin{aligned} g_{\mu\mu}(e_{\hat{0}})^\mu (e_{\hat{0}})^\mu &= -1, \\ g_{\mu\mu}(e_{\hat{i}})^\mu (e_{\hat{i}})^\mu &= 1. \end{aligned} \quad (10.50)$$

Note that for a diagonal metric we also have¹³ $(e_\mu)^{\hat{\mu}} = 1/(e_{\hat{\mu}})^\mu$.

¹³In expressions like these, involving an index μ both with and without a hat, no summation is implied.

The normalization procedure in the last example makes identifying the vielbein components for the orthonormal frame trivial for the diagonal metric. We simply normalize by writing

$$(\mathbf{e}_{\hat{\mu}})^{\mu} = 1/\sqrt{|g_{\mu\mu}|}, \quad (\mathbf{e}_{\mu})^{\hat{\mu}} = \sqrt{|g_{\mu\mu}|}. \quad (10.51)$$

In this way, we can think of the vielbein components as the square roots of the metric components.

Example 10.10

As we shall see later, a spherically symmetric gravitating object of mass M gives rise to the **Schwarzschild metric** with line element given by¹⁴

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (10.52)$$

We can identify an orthonormal frame in this so-called Schwarzschild geometry, which has coordinates ordered (t, r, θ, ϕ) . An observer at rest in this geometry has a velocity vector $\mathbf{u}$ with components

$$u^{\mu} = \left[\left(1 - \frac{2M}{r}\right)^{-\frac{1}{2}}, 0, 0, 0 \right], \quad (10.53)$$

so that $\mathbf{u} \cdot \mathbf{u} = g_{tt} \left(1 - \frac{2M}{r}\right)^{-1} = -1$, as it must. We set $\mathbf{e}_{\hat{t}} = \mathbf{u}$. We then choose

$$\mathbf{e}_{\hat{t}} = \left(1 - \frac{2M}{r}\right)^{-\frac{1}{2}} \mathbf{e}_t, \quad \mathbf{e}_{\hat{r}} = \left(1 - \frac{2M}{r}\right)^{\frac{1}{2}} \mathbf{e}_r, \quad \mathbf{e}_{\hat{\theta}} = \frac{1}{r} \mathbf{e}_{\theta}, \quad \mathbf{e}_{\hat{\phi}} = \frac{1}{r \sin \theta} \mathbf{e}_{\phi}. \quad (10.54)$$

It's not hard to see that these must obey the defining rules above. Alternatively, writing non-zero components of the vielbein explicitly¹⁵

$$(\mathbf{e}_{\hat{t}})^t = \left(1 - \frac{2M}{r}\right)^{-\frac{1}{2}}, \quad (\mathbf{e}_{\hat{r}})^r = \left(1 - \frac{2M}{r}\right)^{\frac{1}{2}}, \quad (\mathbf{e}_{\hat{\theta}})^{\theta} = \frac{1}{r}, \quad (\mathbf{e}_{\hat{\phi}})^{\phi} = \frac{1}{r \sin \theta}. \quad (10.55)$$

Of course the point of identifying vielbein components is to use them, in ways such as that shown in the next example.

Example 10.11

Spacetime is described by a metric line element¹⁶

$$ds^2 = -dt^2 + d\chi^2 + \chi^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (10.56)$$

In the coordinate frame, the energy-momentum tensor for the fluid has components

$$T_{tt} = \rho, \quad T_{\chi\chi} = p, \quad T_{\theta\theta} = p\chi^2, \quad T_{\phi\phi} = p\chi^2 \sin^2 \theta, \quad (10.57)$$

where ρ is an energy density and p is a pressure. We can express these in the orthonormal frame. So, for example, $T_{\theta\theta}$ becomes

$$T_{\hat{\theta}\hat{\theta}} = (\mathbf{e}_{\hat{\theta}})^{\theta} (\mathbf{e}_{\hat{\theta}})^{\theta} T_{\theta\theta} = p. \quad (10.58)$$

We find that in the orthonormal frame, we have

$$T_{\hat{t}\hat{t}} = \rho, \quad T_{\hat{\chi}\hat{\chi}} = p, \quad T_{\hat{\theta}\hat{\theta}} = p, \quad T_{\hat{\phi}\hat{\phi}} = p, \quad (10.59)$$

which is a useful simplification.

↪ **The Schwarzschild metric will be introduced in Chapter 21.**

¹⁴We are here using units in which $G = 1$. The metric is diagonal and has components

$$\begin{aligned} g_{tt} &= -\left(1 - \frac{2M}{r}\right), \\ g_{rr} &= \left(1 - \frac{2M}{r}\right)^{-1}, \\ g_{\theta\theta} &= r^2, \\ g_{\phi\phi} &= r^2 \sin^2 \theta. \end{aligned}$$

¹⁵Since the metric is diagonal, we have

$$\begin{aligned} (\mathbf{e}_{\hat{t}})^{\hat{t}} &= \left(1 - \frac{2M}{r}\right)^{\frac{1}{2}}, \\ (\mathbf{e}_{\hat{r}})^{\hat{r}} &= \left(1 - \frac{2M}{r}\right)^{-\frac{1}{2}}, \\ (\mathbf{e}_{\hat{\theta}})^{\hat{\theta}} &= r, \\ (\mathbf{e}_{\hat{\phi}})^{\hat{\phi}} &= r \sin \theta. \end{aligned}$$

¹⁶This metric implies that the vielbein is diagonal, with components given by the components

$$\begin{aligned} (\mathbf{e}_{\hat{t}})^{\hat{t}} &= 1, & (\mathbf{e}_{\hat{\chi}})^{\hat{\chi}} &= 1, \\ (\mathbf{e}_{\hat{\theta}})^{\hat{\theta}} &= \chi, & (\mathbf{e}_{\hat{\phi}})^{\hat{\phi}} &= \chi \sin \theta. \end{aligned}$$

and

$$\begin{aligned} (\mathbf{e}_{\hat{t}})^t &= 1, & (\mathbf{e}_{\hat{\chi}})^{\chi} &= 1, \\ (\mathbf{e}_{\hat{\theta}})^{\theta} &= \frac{1}{\chi}, & (\mathbf{e}_{\hat{\phi}})^{\phi} &= \frac{1}{\chi \sin \theta}. \end{aligned}$$

10.4 Freely falling frames

There are several possible orthonormal frames that can be identified. In Chapter 6, we discussed the possibility of finding locally inertial frames (LIFs) in which, in addition to the metric being identical to the Minkowski metric, the frame also has the property that the first derivatives of the components $\partial g_{\mu\nu}/\partial x^\alpha$ vanish at the point considered. This implies that the connection coefficients $\Gamma^\mu_{\alpha\beta}$ also vanish at that point.

There are a few methods for identifying LIFs, but one of the most useful is the **freely falling frame**. Recall that a body that is freely falling follows a (timelike) geodesic curve in spacetime. The equivalence principle tells us that a sufficiently small laboratory in free fall should not be able to detect any gravitation. As a result of this, we might expect the laboratory's coordinate system has vanishing connection coefficients. This is indeed the case.

Freely falling frames are therefore defined to possess a system of coordinates in which the connection coefficients vanish along the geodesic that describes their free fall. The frame is described via a set of orthonormal basis vectors $e_{\hat{\alpha}}(\tau)$ that we should determine in order to be able to understand the results of measurements.

Consider the geodesic of the falling observer with proper time τ , which is the curve $x^\mu(\tau)$. The observer's four velocity is $u(\tau) = \frac{dx^\mu}{d\tau} e_\mu$. This vector is identified with the zeroth basis vector $e_{\hat{0}}(\tau) = u(\tau)$. We can find the spatial basis vectors at some point along the geodesic by identifying a set of orthonormal vectors that are perpendicular to u . The basis vectors at other points could be found by parallel transporting the basis vectors along the geodesic. Since the connection coefficients vanish, the freely falling frame is then defined by¹⁷

$$\nabla_u e_{\hat{\alpha}} = 0, \quad (10.60)$$

for all α . Notice that this is automatically satisfied for $e_{\hat{0}} = u$ by definition for a geodesic.

¹⁷We can also identify orthonormal basis 1-forms ω^α with the property $\nabla_u \omega^{\hat{\alpha}} = 0$.

Example 10.12

We shall see in Chapter 22 that falling radially inwards from rest at infinity in the Schwarzschild geometry, a particle has velocity u with components in the coordinate frame of

$$u^\alpha = \left[\left(1 - \frac{2M}{r}\right)^{-1}, -\left(\frac{2M}{r}\right)^{\frac{1}{2}}, 0, 0 \right]. \quad (10.61)$$

We write down that $e_{\hat{t}} = u(\tau)$. As far as it's possible to identify them, diagonal vielbein components are most simple to use. So, as in the previous examples, we write $(e_{\hat{t}})^\theta = 1/r$ and $(e_{\hat{\phi}})^\phi = 1/(r \sin \theta)$. For $(e_{\hat{r}})^\alpha$ we have $g_{\mu\nu}(e_{\hat{r}})^\mu (e_{\hat{r}})^\nu = \eta_{\hat{r}\hat{r}}$ or

$$-\left(1 - \frac{2M}{r}\right) (e_{\hat{r}})^t (e_{\hat{r}})^t + \left(1 - \frac{2M}{r}\right)^{-1} (e_{\hat{r}})^r (e_{\hat{r}})^r = 1. \quad (10.62)$$

Choose $(e_{\hat{r}})^r = 1$, and conclude that the result is

$$\begin{aligned}(e_{\hat{t}})^\alpha &= (e_{\hat{0}})^\alpha = \left((1 - 2M/r)^{-1}, -(2M/r)^{\frac{1}{2}}, 0, 0 \right), \\ (e_{\hat{r}})^\alpha &= (e_{\hat{1}})^\alpha = \left(-(2M/r)^{\frac{1}{2}} (1 - 2M/r)^{-1}, 1, 0, 0 \right), \\ (e_{\hat{\theta}})^\alpha &= (e_{\hat{2}})^\alpha = (0, 0, 1/r, 0), \\ (e_{\hat{\phi}})^\alpha &= (e_{\hat{3}})^\alpha = (0, 0, 0, (r \sin \theta)^{-1}).\end{aligned}\tag{10.63}$$

We saw in Chapter 7, Example 7.10, that for a system in which the connection coefficients vanish, we should have $D\chi/d\tau = d\chi/d\tau$. We should check that χ in a freely falling frame (defined by $\nabla_{\mathbf{u}} e_{\hat{\alpha}} = 0$) has this property.

Example 10.13

Note that in the freely falling frame we have the defining fact that the basis vector and basis 1-forms are parallel transported.

$$\nabla_{\mathbf{u}} e_{\hat{\alpha}} = 0, \quad \nabla_{\mathbf{u}} \omega^{\hat{\alpha}} = 0.\tag{10.64}$$

Our first task is to express the covariant derivative vector $\nabla_{\mathbf{u}} \chi$ in the freely falling frame. To do this we use a vielbein $(e_{\alpha})^{\hat{\mu}}$ to bring its components into the orthonormal freely falling system. The components are $(\nabla_{\mathbf{u}} \chi)^{\hat{\mu}} = (e_{\alpha})^{\hat{\mu}} (\nabla_{\mathbf{u}} \chi)^{\alpha}$, which can be rewritten as¹⁸

$$(e_{\alpha})^{\hat{\mu}} (\nabla_{\mathbf{u}} \chi)^{\alpha} = \nabla_{\mathbf{u}} \left((e_{\alpha})^{\hat{\mu}} \chi^{\alpha} \right).\tag{10.68}$$

But this is just the covariant derivative of the $\hat{\mu}$ component of χ , or $\nabla_{\mathbf{u}} (\chi^{\hat{\mu}})$. The covariant derivative of a component of a vector is the same as the covariant derivative of some scalar function, which (as we saw in the earlier sidenote) is just the directional derivative, which is to say

$$\nabla_{\mathbf{u}} (\chi^{\hat{\mu}}) = u^{\beta} \frac{\partial}{\partial x^{\beta}} \chi^{\hat{\mu}}\tag{10.69}$$

$$= \frac{dx^{\beta}}{d\tau} \frac{\partial}{\partial x^{\beta}} \chi^{\hat{\mu}} = \frac{d\chi^{\hat{\mu}}}{d\tau},\tag{10.70}$$

where τ is the proper time. Putting things together, we conclude

$$\left(\frac{D\chi}{d\tau} \right)^{\hat{\mu}} \equiv (\nabla_{\mathbf{u}} \chi)^{\hat{\mu}} = \frac{d\chi^{\hat{\mu}}}{d\tau}.\tag{10.71}$$

At various points in the remainder of the book, we will deploy vielbein components in order to efficiently perform certain calculations. In the next chapter, we turn to the long-awaited method of determining the curvature of spacetime.

¹⁸ A useful step in seeing this is to write the vielbein components as $(e_{\alpha})^{\hat{\mu}} = \langle \omega^{\hat{\mu}}, e_{\alpha} \rangle$, so we have

$$\langle \omega^{\hat{\mu}}, e_{\alpha} \rangle (\nabla_{\mathbf{u}} \chi)^{\alpha} = \langle \omega^{\hat{\mu}}, \nabla_{\mathbf{u}} \chi \rangle.\tag{10.65}$$

Next, we note that

$$\nabla_{\mathbf{u}} \langle \omega^{\hat{\mu}}, \chi \rangle = \langle \nabla_{\mathbf{u}} \omega^{\hat{\mu}}, \chi \rangle + \langle \omega^{\hat{\mu}}, \nabla_{\mathbf{u}} \chi \rangle,\tag{10.66}$$

but $\nabla_{\mathbf{u}} \omega^{\hat{\mu}} = 0$ by definition of the freely falling frame. So the right-hand side of eqn 10.65 becomes

$$\nabla_{\mathbf{u}} \langle \omega^{\hat{\mu}}, \chi \rangle = \nabla_{\mathbf{u}} \chi^{\hat{\mu}}.\tag{10.67}$$

Chapter summary

- Measurements in general relativity are carried out by observers who carry around an orthonormal coordinate system with basis vectors $e_{\hat{\alpha}}$. An observer with velocity $\mathbf{u}$ has $e_{\hat{0}} = \mathbf{u}$.
- A vielbein, with components $(e_{\mu})^{\hat{\alpha}}$, is a matrix that allows the transformation between orthonormal frames and coordinate frames.
- A particularly convenient frame is an orthonormal one where the observer is at rest relative to the coordinate frame. An alternative is the freely falling frame (defined by $\nabla_{\mathbf{u}} e_{\hat{\alpha}} = 0$ for all α) in which the connection coefficients also vanish.
- In the (stationary) orthonormal frame, the observer's 3-velocity vanishes and so, using $\mathbf{u} \cdot \mathbf{u} = g_{00}(u^0)^2 = -1$, we have $u^0 = (-g_{00})^{-\frac{1}{2}}$. The observer orients their orthonormal frame with $e_{\hat{0}} = \mathbf{u}$, so $e_{\hat{0}} = (-g_{00})^{-\frac{1}{2}} e_0$, or $(e_{\hat{0}})^0 = (-g_{00})^{-\frac{1}{2}}$.
- In the orthonormal frame for a diagonal metric, we have $(e_0)^{\hat{0}} = (-g_{00})^{\frac{1}{2}}$ and $(e_i)^{\hat{i}} = g_{ii}^{\frac{1}{2}}$.

Exercises

- (10.1) Consider spacetime with a line element

$$ds^2 = -dt^2 + a(t)^2 (d\chi^2 + \sinh^2 \chi d\theta^2 + \sinh^2 \chi \sin^2 \phi d\phi^2). \quad (10.72)$$

(a) Using a coordinate system (t, χ, θ, ϕ) , a vector has components in the coordinate frame if $V^{\mu} = (V^t, V^{\chi}, V^{\theta}, V^{\phi})$. What are the vector's components in the orthonormal frame?

(b) A $(1, 2)$ tensor has a non-zero component $G^{\theta}_{\chi\phi}$. What does this become in the orthonormal frame?

- (10.2) (a) Working in the orthonormal frame, find the connection coefficients for flat space represented in cylindrical polar coordinates.

Hint: Remember that the connection coefficients do not transform like tensors, so you cannot simply use the vielbein. You can compute the coefficients directly from the definitions of the basis vectors, or transform using eqn 7.11.

(b) Show that the connection coefficients you have derived obey the rule $\Gamma^{\hat{\mu}}_{\hat{\alpha}\hat{\beta}} - \Gamma^{\hat{\mu}}_{\hat{\beta}\hat{\alpha}} = \langle \omega^{\hat{\mu}}, [e_{\hat{\alpha}}, e_{\hat{\beta}}] \rangle$.

- (10.3) *This problem combines several ideas from the last few chapters and is a useful warm up for some of the physics in Part IV of the book.*

A particle travels radially in a static, spherically symmetric gravitational field described by diagonal metric components $g_{\mu\nu}$ and a velocity vector $\mathbf{u}$.

(a) If the timelike component of the particle's velocity 1-form is given in the static frame of the potential by a constant $u_t = a$, give the other components in terms of a and the components of the metric.

(b) Compute the coordinate velocity dr/dt .

(c) What is the coordinate velocity, as measured by a local observer?

- (10.4) Consider flat spacetime expressed in an orthogonal coordinate system (x^1, x^2, x^3) with a diagonal metric.

(a) Show that the gradient operator acting on a function f becomes

$$\frac{\partial f}{\partial x^{\hat{\mu}}} = \left(\frac{1}{(g_{11})^{\frac{1}{2}}} \frac{\partial f}{\partial x^1}, \frac{1}{(g_{22})^{\frac{1}{2}}} \frac{\partial f}{\partial x^2}, \frac{1}{(g_{33})^{\frac{1}{2}}} \frac{\partial f}{\partial x^3} \right). \quad (10.73)$$

(b) Prove that, in the coordinate frame with this diagonal metric, the divergence can be rewritten as

$$(\nabla_\mu \mathbf{v})^\mu = \frac{\partial v^\mu}{\partial x^\mu} + \frac{1}{\sqrt{g}} \frac{\partial \sqrt{g}}{\partial x^\mu} v^\mu. \quad (10.74) \quad (10.6)$$

(c) Use the result from (b) to show that, expressed in terms of the components in the orthonormal frame, the divergence can be written as

$$\begin{aligned} \nabla \cdot \mathbf{v} = & \frac{1}{\sqrt{g}} \left[\frac{\partial}{\partial x^1} \left(\sqrt{g_{22}g_{33}} v^{\hat{1}} \right) \right. \\ & + \frac{\partial}{\partial x^2} \left(\sqrt{g_{33}g_{11}} v^{\hat{2}} \right) \\ & \left. + \frac{\partial}{\partial x^3} \left(\sqrt{g_{11}g_{22}} v^{\hat{3}} \right) \right]. \end{aligned} \quad (10.75)$$

What does this formula yield for (d) orthonormal cylindrical polar coordinates and (e) orthonormal spherical polar coordinates?

- (10.5) A light signal is emitted by a source on the rim of a centrifuge and detected by a detector at another point on the rim, separated by an angle α . Use the metric for the rotating frame

$$\begin{aligned} ds^2 = & - (1 - \Omega^2 r^2) dt^2 + dr^2 + r^2 d\theta^2 \\ & + dz^2 + 2\Omega r^2 d\theta dt, \end{aligned} \quad (10.76) \quad (10.7)$$

to show that there is no shift in the frequency of the signal.

An alternative to orthonormal local basis vectors was suggested by Newman and Penrose. They considered a pair of real null vectors $\mathbf{l}$ and $\mathbf{n}$ and a pair of complex-conjugate null vectors $\mathbf{m}$ and $\bar{\mathbf{m}}$ obeying

$$\mathbf{l} \cdot \mathbf{m} = \mathbf{l} \cdot \bar{\mathbf{m}} = \mathbf{n} \cdot \mathbf{m} = \mathbf{n} \cdot \bar{\mathbf{m}} = 0. \quad (10.77)$$

The vectors are normalized according to $\mathbf{l} \cdot \mathbf{n} = 1$ and $\mathbf{m} \cdot \bar{\mathbf{m}} = -1$.

(a) If we take the local basis to be

$$\begin{aligned} \mathbf{e}_1 &= \mathbf{l}, & \mathbf{e}_2 &= \mathbf{n}, \\ \mathbf{e}_3 &= \mathbf{m}, & \mathbf{e}_4 &= \bar{\mathbf{m}}, \end{aligned} \quad (10.78)$$

find the components of the local metric $\eta_{\hat{\mu}\hat{\nu}}$.

(b) Find the local basis 1-forms $\omega^{\hat{\mu}}$, assuming the usual relationship $\langle \omega^{\hat{\mu}}, \mathbf{e}_{\hat{\nu}} \rangle = \delta^{\hat{\mu}}_{\hat{\nu}}$.

Suggest vielbein components for a (1+1)-dimensional metric with line element $ds^2 = -dudv$.